Neha Mohan Kumar

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PROFESSIONAL EXPERIENCE

Rutgers University, NJ, USA — **Graduate Research Assistant, 09/24 - 05/25**

- As part of a 2 semester length thesis, I'll be studying Interpretable AI vs Explainable AI and focusing on building afully Interpretable AI systems for critical systems, such as disease diagnosis classifiers.

Gupshup Inc., Bangalore, India — ***Senior Software Engineer (AI Team), 01/23-08/23***

- Headed a team of 2 junior software engineers, building a multi-context Question Answering using a cascade of fine-tuned T5 models and domain specific Flan LLMs, improving the accuracy from 80% to 90%.
- Implemented an indexing method using quantization to index 1TB data, reducing query time to less than 0.1ms with minimal trade-off to accuracy.
- Managed DevOps, including automating the management of Kubernetes pods and services, optimizing pod restart time, saving storage space for large AI models, etc, reducing customer onboarding time by 50%.

Asksid Technology Solutions Pvt Ltd, Bangalore, India — ***Data Scientist, 09/19-12/22***

- Designed a one-click onboarding tool, reducing onboarding time from 1 week to 10 minutes for a 300-product market.
- Built a Conversational AI and an Analytics tool to extract user emotions, demands, and product preferences using Intent Classification, Topic Classification, Sentiment Analysis, and Entity Extraction, with accuracy of 90% (English) and 85% (non-English).
- Implemented a Knowledge Base-backed AI product search engine, increasing search accuracy to 94%.
- Constructed dashboards for self-training and management of Conversation AI engines, including dataset augmentation, model parameter tuning, model training and testing, using ReactJS, reducing customer service demand by 40%.

Graphene Health Tech Pvt Ltd, Bangalore, India — ***AI Engineer, 08/18-08/19***

- Created semi-supervised Aspect Extraction models and Aspect-Based Sentiment Analysis models using BiLSTMs, achieving an accuracy of 83%.
- Designed Topic modeling systems, while heading a group of interns, to extract insights from 10 to 60 GBs of raw consumer reviews.
- Developed a sentiment polarity model for measuring sentiment intensity in natural language sentences ranging from -1 to 1.
- Developed a Neural Network-based Diabetic retinopathy diagnosis system with a test-accuracy of 99.5%.

EDUCATION

Master of Science in Computer Science - **Rutgers University - NJ, USA** — **09/23-05/25**

Bachelor of Technology in Computer Science and Engineering - **National Institute of Technology Karnataka - India** — **08/14-05/18**

PROJECTS

- **Protein sequence localization** - Fine-tuned Facebook's ESM-1b transformer protein language model to classify the location of a given protein sequence, attaining an accuracy of 78%.

PUBLICATIONS

- Prompt Chaining-Assisted Malware Detection: A Hybrid Approach Utilizing Fine-Tuned LLMs and Domain Knowledge-Enriched Cybersecurity Knowledge Graphs - Paper has been accepted as a short paper for the 2024 IEEE International Conference on Big Data. We built an explainable Malware classifier which classifies memory dump data and network packets with the help of a fine tuned Llama model and demonstrated the use of knowledge graphs to keep the LLM grounded and reduce hallucinations, achieving an accuracy of 95%.
- Knowledge-infused and Explainable Malware Forensics - An explainable LightGBM malware classifier, infused with external knowledge through graph embeddings (node2vec), giving an accuracy of 93%.
- Multi-modal Machine Learning Model for Mobile Malware Classification - Multimodal deep learning Android Malware classification framework classifying grayscale image representations of malware and raw numerical features extracted
- Automated Evaluation of Attendance and Cumulative Feedback using Face Recognition - Involved facial recognition using eigenfaces and emotion recognition using haar cascades and Local Binary Patterns Histograms

SKILLS

Python, R, C | LLM, Neural Network, Machine Learning, Data Science, Pytorch, Tensorflow, Keras | MySQL, MongoDB, SQL Server, Elastic Search, Big Query, Neo4j, Allegrograph | Azure Cloud Platform, Google Cloud Platform, AWS | AngularJs, ReactJs, HTML, CSS, Bootstrap, WordPress